

Modul Internet of Things: Menggunakan STM32 + ESP01
Praktikum Senior Project, Lab Dasar 2
Departemen Teknik Elektro dan Teknologi Informasi, Fakultas Teknik,
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Pengaturan IOC pada STM32CubeIDE

1. Buat New Project sesuai dengan tipe STM32 kalian, contoh: STM32F401CDU6, sampai selesai dan tertampil file IOC kalian.
2. Klik “System Core” > RCC > High Speed Clock (HSE) set sebagai “Crystal/Ceramic Resonator”
3. Klik “System Core” > SYS > Debug set sebagai “Serial Wire”
4. Klik “Connectivity” > USART1 > Mode set sebagai “Asynchronous” dan pastikan pada Parameter Settings > Basic Parameters > Baud Rate > 115200 Bits/s
5. Klik save sampai ter-generate main.c, kemudian edit main.c berikut:

Percobaan 1: STM32+ESP01 berlaku sebagai server yang membuat its own wifi connection, dan HP/laptop terkoneksi ke wifi stm32 dengan memasukkan SSID dan Password, kemudian membuka website 192.168.51.1 dan mengendalikan LED STM32 menggunakan tombol di website tersebut.

Di user code begin 2:

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```
uint8_t rxBuffer[512] = {0};
uint8_t ATisOK;
int channel;
int onoff;
int led = 1;
char ATcommand[64];
char ATcommandB[1024];
char ATcommandN[100];
char ATcommandF[100];
char ATcommandT[16];
sprintf(ATcommandB, "<!DOCTYPE html><html>\n<head>\n\
<title>STM32 - ESP8266</title>\n<link href=\"data:image/x-icon;base64,\
A\" rel=\"icon\" type=\"image/x-icon\"><style>\nhtml {\
display: inline-block; margin: 0px auto; text-align: center;}\n\
body{margin-top: 50px;}\n.button {display: block;\n\
width: 70px;\nbackground-color: #008000;\nborder: none;\ncolor: white;\n\
padding: 14px 28px;\ntext-decoration: none;\nfont-size: 24px;\n\
margin: 0px auto 36px; \nborder-radius: 5px;}\n\
.button-on {background-color: #008000;}\n.button-on:active\
{background-color: #008000;}\n.button-off {background-color: #808080;}\n\
.button-off:active {background-color: #808080;}\n\
p {font-size: 14px;color: #808080;margin-bottom: 20px;}\n\
</style>\n</head>\n<body>\n<h1>STM32 - ESP8266</h1>");
sprintf(ATcommandN, "<p>Light is currently on\
</p><a class=\"button button-off\" href=\"/lightoff\">OFF</a>");
sprintf(ATcommandF, "<p>Light is currently off\
</p><a class=\"button button-on\" href=\"/lighton\">ON</a>");
sprintf(ATcommandT, "</body></html>");
int countB = strlen(ATcommandB);
int countN = strlen(ATcommandN);
int countF = strlen(ATcommandF);
int countT = strlen(ATcommandT);

sprintf(ATcommand, "AT+RST\r\n");
memset(rxBuffer, 0, sizeof(rxBuffer));
```

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```
HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
HAL_Delay(500);

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CWMODE_CUR=2\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CWSAP_CUR=\"STM32\", \"12345678\", 1, 3, 4, 0\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPAP_CUR=\"192.168.51.1\"\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}
```

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```
ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPMUX=1\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}

ATisOK = 0;
while(!ATisOK){
    sprintf(ATcommand, "AT+CIPSERVER=1,80\r\n");
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
    if(strstr((char *)rxBuffer, "OK")){
        ATisOK = 1;
    }
    HAL_Delay(500);
}
```

Kemudian di dalam while(1)

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```
memset(rxBuffer,0,sizeof(rxBuffer));
HAL_UART_Receive (&huart1, rxBuffer, 512, 1000);
if(strstr((char *)rxBuffer,"+IPD,0")) channel = 0;
else if(strstr((char *)rxBuffer,"+IPD,1")) channel = 1;
else if(strstr((char *)rxBuffer,"+IPD,2")) channel = 2;
else if(strstr((char *)rxBuffer,"+IPD,3")) channel = 3;
else if(strstr((char *)rxBuffer,"+IPD,4")) channel = 4;
else if(strstr((char *)rxBuffer,"+IPD,5")) channel = 5;
else if(strstr((char *)rxBuffer,"+IPD,6")) channel = 6;
else if(strstr((char *)rxBuffer,"+IPD,7")) channel = 7;
else channel = 100;

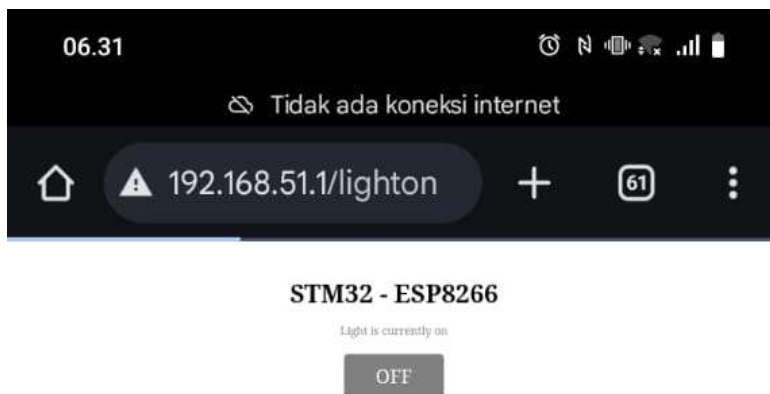
if(strstr((char *)rxBuffer,"GET /lighton")) onoff = 0;
else if(strstr((char *)rxBuffer,"GET /lightoff")) onoff = 1;
else onoff = led;

if(channel<8 && onoff == 1)
{
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_13, GPIO_PIN_SET);
    led = 1;
    sprintf(ATcommand,"AT+CIPSEND=%d,%d\r\n",channel,countB+countF+countT);
    memset(rxBuffer,0,sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1,(uint8_t *)ATcommand,strlen(ATcommand),1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    if(strstr((char *)rxBuffer,">"))
    {
        memset(rxBuffer,0,sizeof(rxBuffer));
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandB,countB,1000);
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandF,countF,1000);
        HAL_UART_Transmit(&huart1,(uint8_t *)ATcommandT,countT,1000);
        HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    }
    sprintf(ATcommand,"AT+CIPCLOSE=%d\r\n",channel);
```


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```
memset(rxBuffer, 0, sizeof(rxBuffer));
HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
channel=100;
}
else if(channel<8 && onoff == 0)
{
    HAL_GPIO_WritePin(GPIOC, GPIO_PIN_13, GPIO_PIN_RESET);
    led = 0;
    sprintf(ATcommand, "AT+CIPSEND=%d,%d\r\n", channel, countB+countN+countT);
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    if(strstr((char *)rxBuffer, ">"))
    {
        memset(rxBuffer, 0, sizeof(rxBuffer));
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandB, countB, 1000);
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandN, countN, 1000);
        HAL_UART_Transmit(&huart1, (uint8_t *)ATcommandT, countT, 1000);
        HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    }
    sprintf(ATcommand, "AT+CIPCLOSE=%d\r\n", channel);
    memset(rxBuffer, 0, sizeof(rxBuffer));
    HAL_UART_Transmit(&huart1, (uint8_t *)ATcommand, strlen(ATcommand), 1000);
    HAL_UART_Receive (&huart1, rxBuffer, 512, 100);
    channel=100;
}
```

Koneksikan wifi hp/laptop Anda ke wifi dengan SSID: STM32 dan password 12345678 dan buka browser dan search link 192.168.51.1 sehingga akan tampil halaman web berikut:



Klik button ON agar LED di STM32 menyala, dan klik OFF agar LED di STM32 padam.

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Percobaan 2: STM32+ESP01 berlaku sebagai client terkoneksi ke Tethering Internet HP, kemudian publish data ke broker test.mosquitto.org, sedangkan Komputer/laptop berlaku sebagai client 2 dan subscribe data dari broker, dengan topic yang sama dengan yang si-publish oleh STM32+ESP01.

```
23 /* USER CODE BEGIN Includes */
24 #include "string.h"
25 #include "stdio.h"
26 /* USER CODE END Includes */
```

Kemudian di antara /* USER CODE BEGIN 0*/ dan /* USER CODE END 0 */

```
81  /* USER CODE BEGIN Init */
82  void send_AT_command(const char *cmd, uint32_t delay_ms) {
83      HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
84      HAL_Delay(delay_ms);
85  }
86
87  void send_and_receive(const char *cmd, uint32_t delay_ms) {
88      uint8_t buffer[512] = {0};
89
90      HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
91      HAL_Delay(delay_ms);
92
93      HAL_UART_Receive(&huart1, buffer, sizeof(buffer), 1000);
94      printf("%s\n", buffer); // Display received response on serial terminal
95  }
96
97  void ESP01_ConnectWiFi(const char *ssid, const char *password) {
98      send_and_receive("AT+RST\r\n", 3000); // Reset ESP
99      HAL_Delay(2000); // Wait for boot
100
101      send_and_receive("AT+CWMODE=1\r\n", 1000); // Set mode: Station (client)
102
103      char wifiCommand[128];
104      snprintf(wifiCommand, sizeof(wifiCommand),
105              "AT+CWJAP=\"%s\",\"%s\"\r\n", ssid, password);
106
107      send_and_receive(wifiCommand, 8000); // Connect to Wi-Fi network
108  }
```

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```
109 void send_AT(const char *cmd, uint32_t delay_ms) {
110     HAL_UART_Transmit(&huart1, (uint8_t *)cmd, strlen(cmd), HAL_MAX_DELAY);
111     HAL_Delay(delay_ms);
112 }
113
114 void send_data(uint8_t *data, uint16_t len) {
115     HAL_UART_Transmit(&huart1, data, len, HAL_MAX_DELAY);
116     HAL_Delay(500); // small delay after sending data
117 }
118
119 void ESP01_ConnectMQTTBroker(const char *broker, uint16_t port) {
120     char cmd[128];
121     snprintf(cmd, sizeof(cmd),
122              "AT+CIPSTART=\"TCP\", \"%s\", %d\r\n", broker, port);
123     send_AT(cmd, 5000);
124 }
125
126 void ESP01_Send_MQTT_CONNECT(const char *clientID) {
127     uint8_t packet[128];
128     uint8_t clientID_len = strlen(clientID);
129     uint8_t total_len = 10 + 2 + clientID_len;
130
131     uint8_t index = 0;
132     packet[index++] = 0x10; // CONNECT command
133     packet[index++] = total_len; // Remaining length
134     packet[index++] = 0x00; packet[index++] = 0x04; // Length of "MQTT"
135     packet[index++] = 'M'; packet[index++] = 'Q'; packet[index++] = 'T'; packet[index++] = 'T';
136     packet[index++] = 0x04; // Protocol level 4
137     packet[index++] = 0x02; // Clean session
138     packet[index++] = 0x00; packet[index++] = 0x3C; // Keep alive = 60 sec
139     packet[index++] = 0x00; packet[index++] = clientID_len;
140     memcpy(&packet[index], clientID, clientID_len);
141     index += clientID_len;
142
143     // Send CIPSEND first
144     char sendCmd[32];
145     snprintf(sendCmd, sizeof(sendCmd),
146              "AT+CIPSEND=%d\r\n", index);
147     send_AT(sendCmd, 1000);
148 }
```


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```
148
149     // Send packet data
150     send_data(packet, index);
151 }
152
153 void ESP01_Send_MQTT_PUBLISH(const char *topic, const char *message) {
154     uint8_t packet[256];
155     uint8_t topic_len = strlen(topic);
156     uint8_t message_len = strlen(message);
157
158     uint8_t remaining_len = 2 + topic_len + message_len;
159     uint8_t index = 0;
160
161     packet[index++] = 0x30;           // PUBLISH command
162     packet[index++] = remaining_len; // Remaining length
163     packet[index++] = 0x00;
164     packet[index++] = topic_len;
165     memcpy(&packet[index], topic, topic_len);
166     index += topic_len;
167     memcpy(&packet[index], message, message_len);
168     index += message_len;
169
170     // Send CIPSEND first
171     char sendCmd[32];
172     sprintf(sendCmd, sizeof(sendCmd),
173             "AT+CIPSEND=%d\r\n", index);
174     send_AT(sendCmd, 1000);
175
176     // Send packet data
177     send_data(packet, index);
178 }
179 /* USER CODE END Init */
```

Kemudian di dalam `int main(){}.`

```
192 /* USER CODE BEGIN 2 */
193 HAL_Delay(2000);
194 ESP01_ConnectWiFi("vivo Y28", "oke123456");
195 ESP01_ConnectMQTTBroker("test.mosquitto.org", 1883);
196
197 HAL_Delay(2000);
198
199 ESP01_Send_MQTT_CONNECT("STM32Client");
200 HAL_Delay(2000);
201
202 /* USER CODE END 2 */
203
204 /* USER CODE BEGIN WHILE */
205 while (1)
206 {
207     ESP01_Send_MQTT_PUBLISH("/sudiyo", "temperature=30\r\n");
208 }
209 /* USER CODE END WHILE */
```

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Buka Command Prompt: Windows + R ketik cmd, cd ke Lokasi Anda menginstall mosquito, kemudian ketikkan:
mosquitto_sub -h test.mosquitto.org -t “/topic”

Percobaan 3: STM32+ESP01 berlaku sebagai client men-subscribe topic yang dikirimkan oleh computer/laptop client ke server dengan topik yang sama, dan menampilkan ke Serial Oscilloscope.

Tidak perlu membuat file project baru, cukup, tambahkan code di bawah dari main.c Anda di **Percobaan 2**. Yang menjadikan STM32 Anda melakukan publish sekaligus subscribe topic ke dan dari broker.

Buka file .ioc kemudian Klik “Connectivity” > USART1 > Mode set sebagai “Asynchronous” dan pastikan pada Parameter Settings > Basic Parameters > Baud Rate > 115200 Bits/s

Hubungkan Rx dan Tx dari downloader Anda ke STM32 Anda, kemudian tambahkan code berikut:

```
Di antara /* USER CODE BEGIN 0 */ dan /* USER CODE END 0 */
void ESP01_Send_MQTT_SUBSCRIBE(const char *topic) {
    uint8_t packet[128];
    uint8_t topic_len = strlen(topic);
    uint8_t remaining_len = 2 + 2 + topic_len + 1; // Packet ID (2 bytes) + Topic length (2 bytes) + 1

    uint8_t index = 0;
    packet[index++] = 0x82; // Fixed header: SUBSCRIBE packet type (0x80) + flags (0x02)
    packet[index++] = remaining_len; // Remaining Length
    packet[index++] = 0x00; // Packet ID MSB
    packet[index++] = 0x01; // Packet ID LSB
    packet[index++] = 0x00; // Topic length MSB
    packet[index++] = topic_len; // Topic length LSB
    memcpy(&packet[index], topic, topic_len); // Topic
    index += topic_len;
    packet[index++] = 0x00; // QoS 0

    // Tell ESP to prepare for sending
    char sendCmd[32];
    sprintf(sendCmd, "AT+CIPSEND=%d\r\n", index);
    send_AT(sendCmd, 100);

    // Send the packet
    send_data(packet, index);
}
```

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```
char* msgok = ".";
// Receiving and parsing MQTT SUBSCRIBED messages
void loop_receive_and_parse(void) {
    uint8_t rx_buf[256];
    memset(rx_buf, 0, sizeof(rx_buf));
    // char* okey = "+IPD,0,14:humidity=78.92";
    // char* okey2 = strstr((char*)okey, "+IPD,");
    // char* okey3 = strchr(okey2, ':');
    // okey3++;
    // msgok = strchr(okey3, '=');
    // msgok++;
    // float value = 0;
    // sscanf(msgok, "%f", &value);
    // subscribed_value = value;

    if (HAL_UART_Receive(&huart1, rx_buf, sizeof(rx_buf), 100) == HAL_OK) {
        char *start = strstr((char*)rx_buf, "+IPD,");
        if (start) {
            char *payload = strchr(start, ':');
            if (payload) {
                payload++; // Move past ':'
                char *equal_sign = strchr(payload, '=');
                if (equal_sign) {
                    equal_sign++; // Move past '='
                    float value = 0;
                    sscanf(equal_sign, "%f", &value);
                    subscribed_value = value;
                }
            }
        }
    }
}
```

Di dalam int main() {}

```
ESP01_Send_MQTT_CONNECT("STM32Client");
HAL_Delay(200);
ESP01_Send_MQTT_PUBLISH("/okep", "temp=30\r\n");
ESP01_Send_MQTT_SUBSCRIBE("/okes");
```

Di dalam while(1) {}

```
loop_receive_and_parse(); // Update subscribed_value
sprintf(msg, "%f\r\n", subscribed_value);
HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg), HAL_MAX_DELAY);
```

Buka Command Prompt: Windows + R ketik cmd, cd ke Lokasi Anda menginstall mosquito, kemudian ketikkan:
mosquitto_pub -h test.mosquitto.org -t "/topic" -m "temp=30"
enter

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Kemudian buka Serial Oscilloscope, dan buka serial monitor dan oscilloscope nya bersamaan. Jika berhasil, maka yang tertampil adalah angka 30.

Jika berhasil, publish data tersebut dengan mengganti nilai temp, menjadi berikut:

```
mosquito_pub -h test.mosquitto.org -t "/topic" -m "temp=35"  
enter  
mosquito_pub -h test.mosquitto.org -t "/topic" -m "temp=40"  
enter  
mosquito_pub -h test.mosquitto.org -t "/topic" -m "temp=45"  
enter
```

Jika output di serial monitor berubah menunjukkan 35, 40, dan 45, maka Anda sukses.

Percobaan 4. Publish data secara kontinyu menggunakan python code, menjadikan computer Anda sebagai client1 sedangkan STM32 Anda men subscribe dan menampilkan data tersebut di serial oscilloscope.

Project yang dipakai sama dengan **Percobaan 3**, Anda hanya butuh membuat file Python Bernama yourname.py berikut:

```
1  import paho.mqtt.client as mqtt  
2  import time  
3  import random  
4  from paho.mqtt.client import MQTTv311 # this is important!  
5  
6  broker_address = "test.mosquitto.org"  
7  topic = "sdr/pwm"  
8  
9  client = mqtt.Client(client_id="PythonPublisher", protocol=MQTTv311, transport="tcp")  
10 client.connect(broker_address)  
11  
12 while True:  
13     data = random.uniform(20.0, 30.0)  
14     client.publish(topic, "temperature="+str(data))  
15     print("temperature="+str(data))  
16     time.sleep(0.1)
```

Eksekusi file python Anda dengan mengetik kan

python yournem.py

di CMD Anda kemudian klik Enter.

Jangan lupa sesuaikan nama topik yang ingin Anda subscribe di STM32 harus sama dengan nama topic yang Anda publish di file python ini.

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Percobaan 5. [Komunikasi dua buah STM32] Pada percobaan ini, pasang Potensiometer dengan kaki Tengah ke pin ADC PA6 di STM32 **Anda**, sedangkan kedua kakinya, satu dihubungkan ke pin 3V3 dan G di STM32 Anda. Gunakan ADC Polling.

Hubungkan kutub positif dan negative kipas ke power supply 5V (merah) dan Nol (hitam). Hubungkan pin PWM kipas ke pin PWM PA7 di STM32 **teman Anda**.

STM32 **Anda**: Buat project baru persis dengan di **Percobaan 2**, dengan mengubah data yang publish adalah data bacaan ADC.

STM32 **teman Anda**: Buat project baru persis dengan **Percobaan 3** (fungsi send_AT, send_data, ESP01_ConnectMQTTBroker, dan ESP01_Send_MQTTConnect disamakan dengan **Percobaan 2**), dengan menambahkan perintah membangkitkan PWM di fungsi loop_receive_and_parse()

Hubungkan STM32 **Anda** dan **teman Anda**, ke satu tethering HP yang sama. Sesuaikan nama topik yang Anda publish dengan topik yang disubscribe teman Anda. Kemudian Run.

Anda dan teman Anda sukses jika dengan memutar potensiometer Anda, kipas di teman Anda ikut berubah kecepatannya.